Start with data → data can be anything from emails to videos

↓

ensure data is tidy → data is organized in a way that it can be ready for analysis

↓

now we can evaluate data using RStudio (an application) = DATA SCIENCE!

↓

modes of evaluating data

Visualizations

- univariate
  - one variable graphs
- multivariate
  - 2+ variables
- bivariate
  - 2 variables
- Point Map
  - plotting locations of individual observations
- Contour Map
  - plotting density/distribution of observations
- Chloropleth Maps
  - plotting outcomes in different regions

numerically

- head() summarizes first 6 rows
- mathmatical functions
  - ↳ can use R-studio like a calculator

MANY MORE FUNCTIONS!

- can change colors of graphs, fills, titles, etc. → customizable = means millions of graph creation options, but it needs to be effective (ethical, correct, and not "ugly")

All have the goal of making the data a story!!

Accessible friendly graphs:

- a concept that completely makes sense though I have not thought about it which is why I chose to research it more

- according to CDC btwn. 2019 + 2025
  36.4% of people have some difficulty w/ vision or hearing not including color blindness
  ↳ lot of people!

- making sure graphs/information can be equally accessed among everyone (especially if the data is meaningful) is crucial

- ways to do this is to
  1. Use labels instead of color when possible (italicize/underline/etc. to show differ.)
  2. Use audio/reader function
  3. Avoid bias
  4. Use a color-blind friendly template